1.Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
d1_96343_yogesh@localhost [sales]> select * from orders where cnum = (select cnum from customers where cname = 'Cisneros') ;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.002 sec)
```

7.
Extract all the orders of Motika.

```
d1_96343_yogesh@localhost [sales]> select * from orders where snum = (select snum from salespeople where sname='Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.001 sec)
```

8.Find all the orders of salespeople servicing customers in London.

```
d1_96343_yogesh@localhost [sales]> select * from orders where cnum IN (select cnum from customers where city ='London' );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.007 sec)
```

6.Write a query that selects all orders for amounts greater than any for the customers in London.

```
d1_96343_yogesh@localhost [sales]> select * from orders where amt>any(select amt from orders where cnum IN (select cnum from customers where city ='London' ));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.001 sec)
```

4.Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
d1_96343_yogesh@localhost [sales]> select * from customers where rating >= any (select rating from customers where snum = (select snum from salespeople where sna
me = 'Serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.001 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
d1_96343_yogesh@localhost [sales]> select * from customers where rating>any(select rating from customers where city = 'Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.001 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’

```
d1_96343_yogesh@localhost [sales]> select * from orders where amt>ANY(select amt from orders where odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.001 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
d1_96343_yogesh@localhost [sales]> select * from orders where  amt<ANY(select amt from orders where cnum in (select cnum from customers where city='San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.001 sec)
```

2.Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
d1_96343_yogesh@localhost [sales]> select o.amt,c.cname,c.rating   from orders o inner join customers c on o.cnum = c.cnum where amt>(select avg(amt) from orders
);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.001 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
d1_96343_yogesh@localhost [sales]> select * from customers where rating>ALL(select rating from customers where city='London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.001 sec)
```

5.Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
d1_96343_yogesh@localhost [sales]> select * from salespeople s where s.city not in (select city from customers c where c.snum = s.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.003 sec)
```

9.Find names and numbers of all salesperson who have more than one customer.

```
d1_96343_yogesh@localhost [sales]> select sname,snum from salespeople where snum in (select snum from customers group by snum having count(snum)>1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.001 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
d1_96343_yogesh@localhost [sales]> select snum,sname,city,(select count(snum) from customers c group by snum having count(snum)>1 and c.snum=s.snum ) as customer
_count from salespeople s where snum in (select snum from customers group by snum having count(snum)>1);
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.001 sec)
```

3.Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
d1_96343_yogesh@localhost [sales]> select snum ,sum(amt) as totalAmount   from orders group by snum having totalAmount>(select max(amt) from orders) ;
+------+-------------+
| snum | totalAmount |
+------+-------------+
| 1001 |    15382.07 |
+------+-------------+
1 row in set (0.001 sec)
```